DSPi: a digital sound processor for the price of a pizza or two

Introduction and build guide: a USB sound card with a parametric equalizer, made from a Raspberry Pi Pico 2 and a PCM5102A converter



The finished DSPi in a 3D-printed enclosure with a sticker (enclosure and sticker are included in the repository).

What you build	a USB sound card with a built-in DSP that sends already-corrected sound to active speakers, an amplifier or a headphone amplifier
Cost	≈ €12–15 for the main parts, roughly €17–20 including cables (ch. 5)
Time	1–2 hours for wiring and setup, another evening for tuning by ear
Skills needed	connect 5 wires following a diagram, copy a file to the Pico, use a PC application; soldering only if your DAC module comes without header pins or jumpers
Status	25 Oct 2026, DSPi firmware v1.1.5 (stable) and v1.1.6-beta5; screenshots from DSPi Console for Windows with a device connected
Enclosure	the 3D-printed enclosure, sticker and source files are in the same repository – see the separate print and assembly guide

1. Introduction: what DSPi is and why you might care

Most cheap active and computer speakers share the same weaknesses: small drivers that cannot reproduce deep bass, a mid-heavy voicing or, on the contrary, exaggerated bass, and crude BASS/TREBLE knobs that lift a wide band including frequencies you do not want to boost. A software equalizer on the computer helps, but it is tied to one computer and one operating system. Ready-made digital signal processors (DSP) do it properly, but they cost hundreds.

DSPi is open-source firmware by Weeb Labs (GPL-3.0 license) that turns a Raspberry Pi Pico microcontroller into a USB sound card with a full-featured DSP. A computer, phone or tablet sees it as an ordinary audio device without drivers; all sound processing runs in the Pico. Add a cheap converter (DAC) to the Pico and feed its output into anything with a line input.

Typical uses

- **Correcting small speakers** – cut the sub-bass the driver cannot play, lift the range it still handles, tame excessive midrange, add loudness for quiet listening. Chapter 9 covers this case.
- **Active crossover** – splitting the signal for woofer and tweeter with separate amplifiers (2 to 4 ways).
- **Adding a subwoofer** – a dedicated subwoofer output with crossover frequency and delay.
- **Headphones** – crossfeed that softens the unnaturally sharp stereo image in headphones.
- **Room correction** – with a measurement microphone you can suppress room resonances.

> **Realistic expectations**
> DSP does not change physics. Even after correction, a small driver will not play deep bass like a subwoofer. The result is a balanced, cleaner and fuller sound within the range the speaker can handle, without needless strain where it cannot.

2. How it works

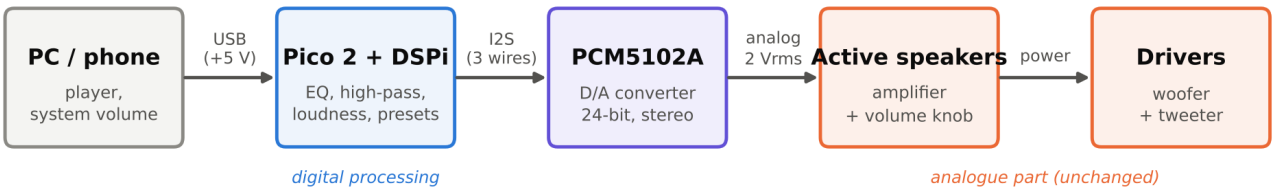


Fig. 1 – Signal chain. The Pico appears to the computer as a sound card; it processes the digital audio, the converter turns it into an analogue signal, which goes to the line input of the speakers or amplifier.

The computer (or phone) sends audio over USB to the Pico. The DSPi firmware processes it and passes it over the **I2S** bus (three wires: bit clock BCK, channel clock LRCK and data DIN) to the PCM5102A converter. The converter produces a 2 Vrms analogue signal that goes through a 3.5 mm jack to the line input. The speakers and amplifier are not modified in any way.

Inside DSPi, audio passes through a fixed chain of stages. These are the important ones for this guide:

Stage	What it does	Where we use it
Preamp	gain / attenuation of the L and R inputs at the start of the chain	headroom for bass boost (ch. 9.2)
Master EQ	10 bands of parametric EQ per channel, 14 filter types	main tone correction (ch. 9.3)
Loudness	volume-dependent correction per ISO 226:2003	fuller sound at low volume (ch. 9.4)
Matrix	routing inputs to outputs, gain and phase inversion	speaker polarity test (ch. 9.1)
Output EQ and crossover	another 10 bands + up to 4 steep crossover filters per output	steep high-pass (ch. 9.3)
Master volume	output level ceiling, −128 to 0 dB	setting the output level (ch. 9.2)

3. What DSPi can do – overview

The firmware evolves quickly. The table summarises the state as of 16 Oct 2026; features marked (beta) are only in test releases so far.

Area	Options
Inputs	USB audio 16/24-bit, 44.1 / 48 / 96 kHz (Windows, macOS, Linux, iOS without drivers); S/PDIF with an optical receiver; I2S, e.g. from a PCM1808 A/D converter for an analogue input
Outputs	up to 4 stereo slots (Pico 2), each S/PDIF or I2S; a separate PDM subwoofer output; optional master clock for converters that need it
Equalization	10 bands of parametric EQ per channel, 14 filter types including the Linkwitz transform; 110 bands in total on the Pico 2
Crossovers	Linkwitz-Riley, Butterworth and Bessel up to 8th order (48 dB/oct), up to 4 per output
Dynamics and perception	loudness per ISO 226:2003, volume leveller, crossfeed for headphones, psychoacoustic bass, subharmonic synthesizer (beta)
Time and routing	input/output matrix with phase inversion, up to 85 ms delay per output
Control	DSPi Console (Windows, macOS, Linux) and DSPi Terminal apps; 10 presets; buttons, potentiometers, encoders, IR remote and display connected directly to the Pico
Diagnostics	peak and clip meters, CPU load, test signal generator, spectrum analyser (beta)

4. Comparison with other solutions

The same goal – better sound from your existing speakers – can be reached in several ways. Prices are indicative, as of October 2026.

Solution	Price	Correction accuracy	No PC app needed	Effort
BASS/TREBLE knobs on the speakers	€0	coarse, fixed bands	yes	none
Analogue tone control module (e.g. NE5532 or LM1036 based)	≈ €7–20	fixed bands, similar to knobs	yes	power supply, often soldering
Graphic EQ on the PC	free	fixed bands, imprecise	no	low
Equalizer APO (parametric EQ on Windows)	free	high	no, Windows only	low
Wondom APM2 (ADAU1701) + ICP5 programmer	≈ €50 + power supply	high	yes	SigmaStudio or app
miniDSP 2x4 HD	≈ €240–260	high, incl. room correction	yes	low, finished box
Pico 2 + PCM5102A + DSPi	**≈ €12–15 + cables**	**high**	**yes**	**connect 5 wires**

Advantages of the DSPi solution

- **Price.** Parametric EQ, steep crossovers, loudness, presets and delay for a fraction of the price of finished processors.
- **Independent of the computer.** The correction runs in the Pico. It sounds the same from Windows, macOS, Linux and iOS; system updates or a changed default device will not break the settings.
- **Precision.** Parametric filters target a specific frequency with a chosen bandwidth. The Pico 2 computes in floating point and uses a numerically more accurate filter topology for low frequencies.
- **Openness.** Source code and documentation are public; the project has an active community.
- **Room to grow.** The same Pico can later drive a subwoofer, an active crossover or your own control panel.

Disadvantages and limitations

- **The input is primarily digital.** The basic setup receives audio over USB. An analogue source (e.g. a TV via jack) needs an extra A/D converter module, an optical S/PDIF output needs a receiver.
- **A young project.** The firmware evolves fast and some features are in beta. Use the stable release for everyday use.
- **A DIY build.** You need to connect wires and possibly bridge solder pads on the DAC module; it is not a finished product.
- **Tuning needs the app.** Settings are made in DSPi Console; afterwards the app does not need to run.
- **Possible hum.** USB connects the computer ground with the speaker ground; with a ground loop a USB isolator helps (ch. 11).

5. Hardware and prices

Item	Where (example)	Price	Note
Raspberry Pi Pico 2 H (RP2350, with soldered headers)	raspberrypi.com	≈ €7	H version = headers soldered; micro-USB connector
PCM5102A DAC module, e.g. GY-PCM5102 (I2S, stereo)	electronics shops, AliExpress	≈ €4–6	2 Vrms stereo output, 3.5 mm jack
Micro-USB data cable	anywhere	≈ €2–6*	must carry data, not just charge
Female–female jumper wires, 5 pcs	anywhere	≈ €1–2*	if the DAC has header pins; otherwise solder
3.5 mm jack – 2× RCA or jack–jack cable	anywhere	≈ €2–4*	depending on the speaker / amplifier input
Optional: USB isolator	anywhere	varies*	only for hum from a ground loop
Optional: enclosure	3D print (files in the repository)	≈ €1 of filament	see the enclosure guide
Total (main parts)		**≈ €12–15**	roughly €17–20 including cables

* indicative price. Shops are only examples; the same parts are sold by many others. Czech prices as of October 2026: Pico 2 H 179 CZK (RPishop.cz), GY-PCM5102 98 CZK (LaskaKit.cz).

Extension for headphones and a control panel (ch. 10)

Item	Price	Note
Second PCM5102A DAC module	≈ €4–6	separate output for headphones
Headphone amplifier (finished module or kit)	varies*	the PCM5102A is not meant to drive headphones directly
3 tactile switches (buttons)	≈ €0.5–1*	preset switching, mute
Rotary encoder module with push button	≈ €1–2*	volume, preset selection
38 kHz IR receiver (3-pin, 3.3 V)	≈ €1–2*	remote control – an old TV remote will do

Why Pico 2 rather than the original Pico

The firmware runs on both, but the Pico 2 (RP2350) has a hardware floating-point unit. The older RP2040 computes in fixed point. The price difference is about a euro.

Parameter	Pico (RP2040)	Pico 2 (RP2350)
Arithmetic	fixed point (Q28)	floating point (FPU)
EQ bands in total	70 (10 per channel)	110 (10 per channel)
S/PDIF or I2S output slots	2 stereo	4 stereo
Sample rates	44.1 / 48 / 96 kHz	44.1 / 48 / 96 kHz

6. Wiring

Before you start
Always wire with the USB cable disconnected. Most important: do not swap 5 V (VBUS) and GND – that can destroy the module.

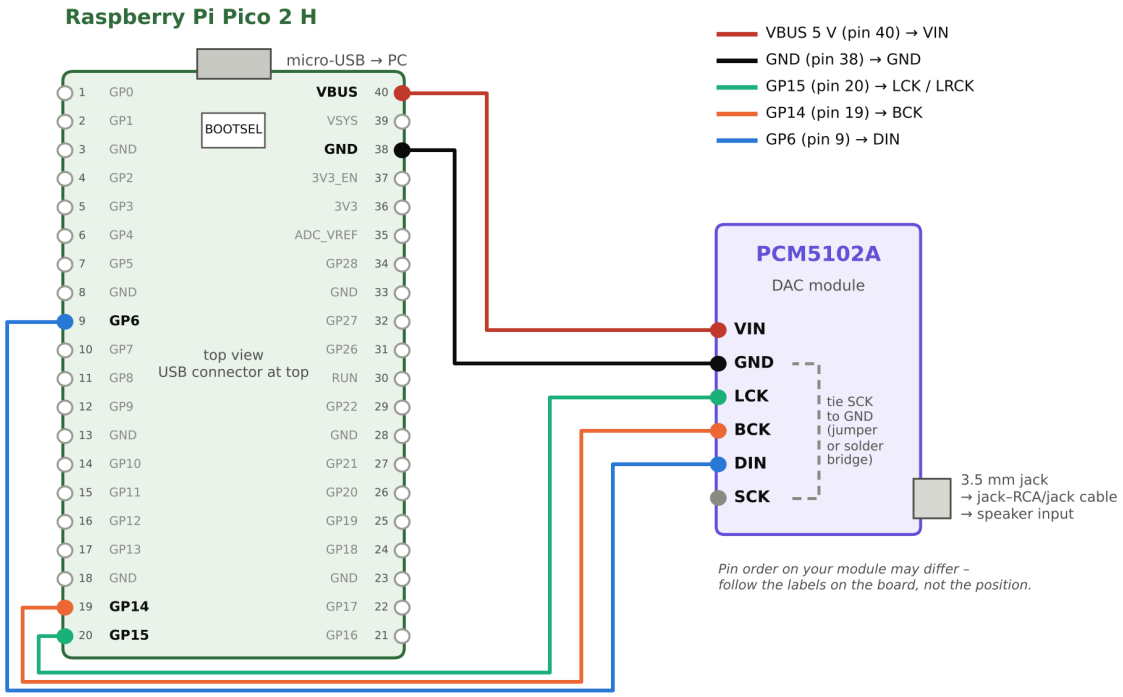


Fig. 2 – Wiring the Pico 2 H and the PCM5102A module. The Pico is drawn from above with the USB connector at the top; numbers next to the pins are physical pin numbers 1–40, GPxx labels are GPIO numbers.

Wire	Pico 2 (pin / function)	PCM5102A module	Purpose
red	pin 40 / VBUS (5 V from USB)	VIN (VCC)	module power
black	pin 38 / GND	GND	common ground
orange	pin 19 / GP14	BCK	I2S bit clock
green	pin 20 / GP15	LCK (LRCK)	left/right channel clock
blue	pin 9 / GP6	DIN	audio data (outputs 1–2)

GP6, GP14 and GP15 are the DSPi defaults for the first output slot and the I2S clocks. LRCK is always on the pin one higher than BCK. The pins can be changed in the app, but there is no reason to.

Configuring the PCM5102A module

The converter has several configuration inputs. On common modules they are brought out as solder bridges on the back (often labelled H1L–H4L) plus a separate SCK pin or bridge. Many modules come configured correctly, but check after delivery:

Input	Setting	Why
SCK (system clock)	to GND	the converter derives its clock from BCK, no MCK signal from the Pico is needed
FMT (H4L)	L (low)	I2S format; DSPi sends standard I2S (Philips): 24 bits in a 32-bit slot

Input	Setting	Why
XSMT (H3L)	H (high)	soft mute off; with L the module stays silent
FLT (H1L)	L	normal filter latency
DEMP (H2L)	L	de-emphasis off

If the seller states different labels or defaults for your module, follow them. The function of the inputs is the same on every PCM5102A module.

Connect the module output (3.5 mm jack) with a cable to the line input (AUX, LINE IN) of the speakers or amplifier.

7. Flashing the firmware

1. On the DSPi releases page download the file for **RP2350**, e.g. `DSPi-RP2350-v1.1.5.uf2` (stable). The RP2040 file is for the original Pico and does not work on the Pico 2.
2. Hold the **BOOTSEL** button on the Pico while plugging the USB cable into the computer. Then release the button.
3. A new drive appears on the computer (usually named `RP2350` for the Pico 2).
4. Drag the .uf2 file onto it. The Pico restarts by itself and the drive disappears.
5. The Pico now appears as the audio device **Weeb Labs DSPi**. It works without drivers on Windows, macOS and Linux.

For later updates you do not need BOOTSEL: in DSPi Console choose ☰ → **Update Firmware…**, the Pico restarts into the same mode and you just drag the new .uf2 file onto the drive. Newer app versions flash the firmware themselves.

8. DSPi Console: installation and first setup

DSPi is configured with the **DSPi Console** application for Windows, macOS and Linux (github.com/WeebLabs); a command-line tool dspictl is also available for scripts. The screenshots in this chapter are from the Windows version with a device connected (they show a real correction of small desktop speakers); the versions for other systems work the same way.

Matching versions
The DSPi Console and firmware versions must match, including the beta tag (firmware 1.1.5 needs Console 1.1.5). With a mismatch the app may hide features or even write values into the wrong parameters.

8.1 Installation (Windows)

1. On the DSPi Console for Windows releases page download `DSPi.Console.v1.1.5.zip` (≈ 31 MB).
2. If you do not have it, install the **.NET 8 Desktop Runtime** from Microsoft. Requirement: 64-bit Windows 10 (1809) or newer.
3. Unpack the archive anywhere and run `DSPiConsole.exe`. The app is portable, nothing is installed.
4. If Windows SmartScreen warns about an unrecognised app (it is not digitally signed), choose *More info* → *Run anyway*.

8.2 Main window



Screenshot 1 – Main window with DSPi connected.



Screenshot 2 – The ☰ menu.

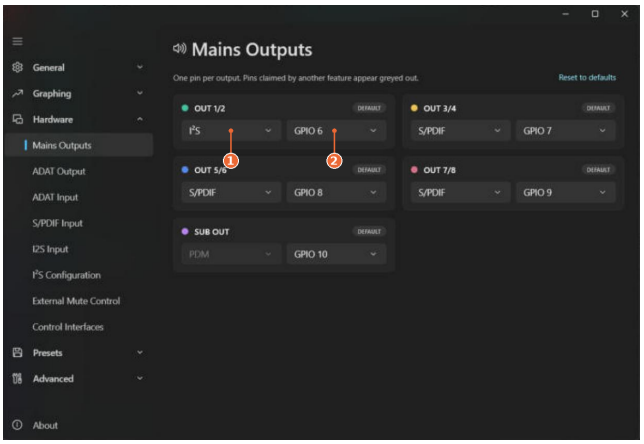
	Element	Purpose
1	Menu ☰	all functions and windows, saving presets (Save Preset, Revert Preset), Save Output Config, Update Firmware
2	Channels	inputs **USB 1** and **USB 2** (left and right channel from the computer) and outputs **I2S 1 L** and **I2S 1 R**. A click opens the channel editor (Screenshots 7 and 8): parametric EQ and **Preamp** for inputs, **PEQ** and **XO** (crossover) tabs for outputs. Click again to go back.
3	Graph	resulting frequency response; filter points can be dragged with the mouse in the channel editor
4	Toolbar	from the left: Matrix Mixer, Settings, Loudness, Crossfeed, Psychoacoustic Bass, Volume Leveller, Stats for nerds, Bypass Master EQ. Left click toggles the function, right click opens its settings.
5	Preset, Source, Master Volume	preset selection (an asterisk * and a **Save** button = unsaved changes), audio source (USB, S/PDIF, I2S) and device volume; the ▾ arrow switches to **User Volume**
6	Device	the connected DSPi with its serial number; *Disconnected* without a device
7	Filter overview	filter tables of all channels; each band shows type (LS = low shelf, PK = peaking, HS = high shelf, LC = low cut), frequency, gain and Q

8.3 Connecting

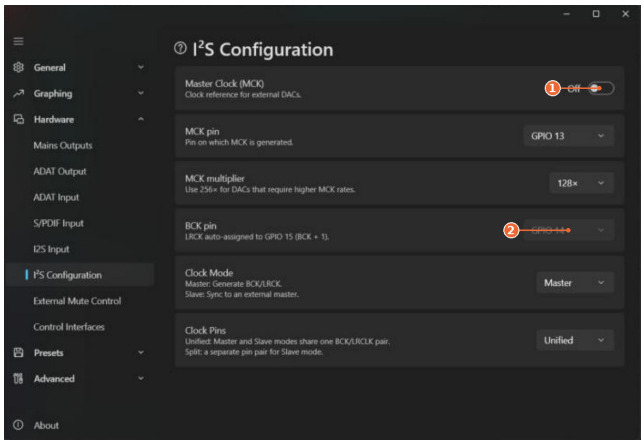
Connect the Pico with the firmware flashed. The device appears at the bottom and the channels appear in the left panel. If it does not connect, check the data cable. If the app reports that libusb sees no device, assign the **WinUSB** driver to the DSPi control interface with Zadig – this only affects control, audio works anyway.

8.4 Switching the output to I²S (required)

The outputs are set to S/PDIF by default. The PCM5102A needs I²S – without this step you get silence. Open ☰ → **Settings…** (or the gear in the toolbar) and in the **Hardware** section:

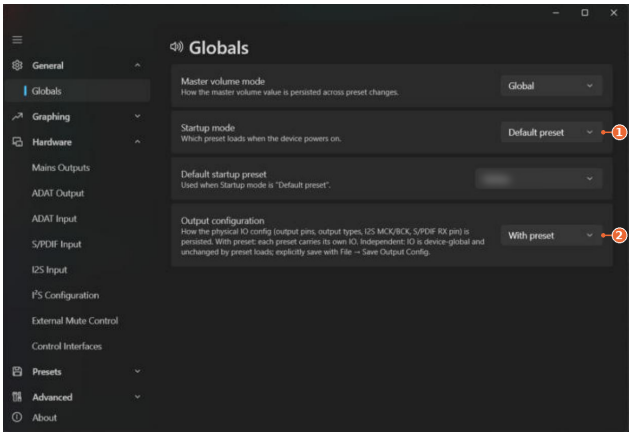


Screenshot 3 – Hardware → Mains Outputs: for OUT 1/2 switch (1) the type S/PDIF → I²S and check (2) pin GPIO 6.



Screenshot 4 – Hardware → I²S Configuration: (1) Master Clock (MCK) off, (2) BCK pin GPIO 14; LRCK is assigned automatically to GPIO 15.

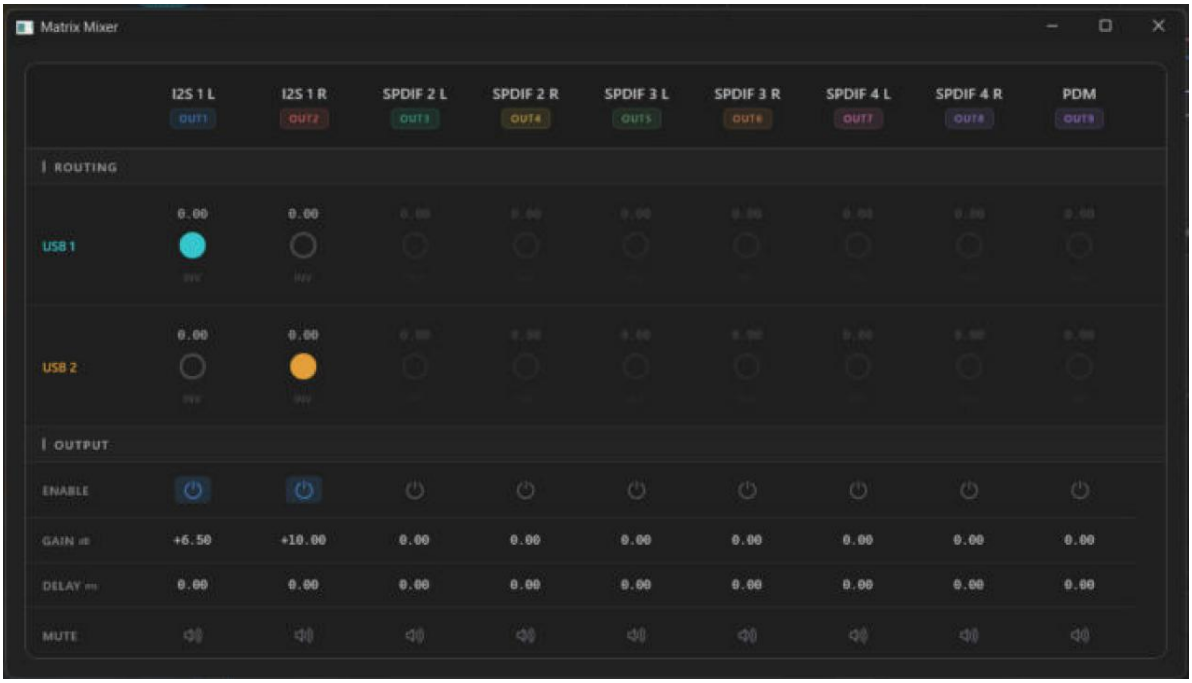
The change of output type applies immediately. **But you must save it separately**: by default the output configuration is stored independently of presets, with ☰ → **Save Output Config**. The mode can be changed in *Settings* → *General* → *Globals* → *Output configuration* (Independent / With preset). An unsaved change reverts to S/PDIF after unplugging. The second output for headphones (OUT 3/4, ch. 10.2) is switched the same way.



Screenshot 5 – General → Globals: (1) Startup mode, (2) Output configuration.

- **Output configuration**: *Independent* = the output configuration is shared by all presets and saved with Save Output Config; *With preset* = each preset carries its own configuration and saves it with Save Preset.
- **Startup mode**: which preset is loaded at power-on (ch. 10.3).

8.5 Finishing



Screenshot 6 – Matrix Mixer: USB 1 → I2S 1 L, USB 2 → I2S 1 R; below: enable, gain, delay and mute for each output.

1. Open the **Matrix Mixer** (☰ → Matrix Mixer…) and check the routing: USB 1 → I2S 1 L, USB 2 → I2S 1 R.
2. In Windows open *Settings* → *System* → *Sound* and choose **Weeb Labs DSPi** as output; in the device properties set the format to 24 bit / 48 kHz.
3. Set the system and speaker volume low, play music and turn it up gradually.
4. Save the settings: ☰ → **Save Preset** (Ctrl+S). Otherwise DSP settings are kept only until power is removed. **Revert Preset** restores the last saved version.

Where to find things in Settings

Section → page	Content
General → Globals	Master volume mode (Global / Per preset), Startup mode (Default preset / Last used), Default startup preset, Output configuration (Independent / With preset)
Hardware → Mains Outputs	type (S/PDIF / I²S) and pin of each output
Hardware → I²S Configuration	MCK and BCK/LRCK clocks
Hardware → External Mute Control	hardware mute of the converter (ch. 10.5)
Hardware → Control Interfaces	control over UART and I2C
Presets → UI	quick-save button for presets with unsaved changes

Some pages only appear with a device connected. Changes written to the device memory are collected in the Settings window and a bar with the number of pending changes and **Save to flash** / **Discard** buttons appears at the bottom.

> **Checkpoint**
> Does it play? Still without correction, purely as a USB sound card. This is a good moment to listen to how your speakers sound unmodified – it lets you judge later what the correction achieved.

9. Tuning the sound

Tuning has a fixed order. Each step builds on the previous one: first make sure the speakers are not playing against each other, then prepare headroom for boost, only then shape the sound and finally adjust volume and loudness. The specific values in this chapter are an example for small desktop speakers; the principles apply generally.

9.1 Speaker polarity test

If one speaker is wired with reversed polarity (it happens with cheap sets with a fixed cable), the speakers play out of phase and the bass cancels out – no equalizer can fix that. DSPi can invert the phase in software:

1. Play a track with strong bass and sit at the listening position.
2. In the **Matrix Mixer** enable **INV** (phase invert) at the crossing of input USB 2 and output I2S 1 R.
3. Compare. Much more bass → the polarity was reversed, leave the inversion on. Less bass or the sound "escapes to the sides" → polarity is fine, turn the inversion off.

9.2 Headroom and volume (gain staging)

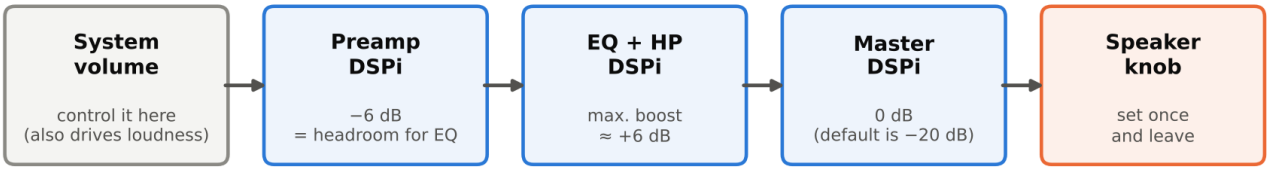


Fig. 3 – Where the level is controlled in the chain and how to set each element.

A digital signal has a fixed ceiling of 0 dBFS. If the music already reaches the ceiling and the equalizer adds +6 dB, the result is clipped and distorts. That is why boost is "prepaid" by attenuating at the input:

- Reduce the **Preamp** by as much as the biggest boost of your correction (−6 dB in the example below; the computed peak is +5.9 dB). You find it in the input channel editor (click USB 1 or USB 2); with Link Pair you adjust both channels at once.
- **Master Volume 0 dB** (slider at the bottom of the main window; the default is −20 dB). The output is then the full 2 Vrms and the speaker knob does not need to be turned up high.
- **Control the volume in the operating system** (Windows, macOS…). The loudness function follows it (ch. 9.4).
- **Set the speaker knob once**: at 100 % system volume, to the loudest level you will ever want. Then leave it alone.

> The clip meters in DSPi Console must stay below the ceiling during the loudest passages. If they light up, increase the attenuation in the Preamp, not elsewhere.

9.3 Tone correction

The procedure is the same for any speakers: find where the driver stops playing (manufacturer data, listening or measurement), cut below that limit, boost moderately just above it and then even out the rest of the range. The following table is a starting point for typical small desktop speakers with 3–4" drivers that sound mid-heavy and lack both bass and treble.

Where in DSPi	Filter type	Frequency	Gain	Q	Purpose
I2S 1 L and R – XO tab	4th-order Butterworth high-pass (24 dB/oct)	50 Hz	—	—	cuts sub-bass the driver cannot play
USB 1/2 – PEQ, band 1	peaking	85 Hz	+6 dB	1.0	bass in the range a small driver can still play
USB 1/2 – PEQ, band 2	peaking	900 Hz	−3.5 dB	0.7	tames the mid-heavy character
USB 1/2 – PEQ, band 3	high shelf	8 kHz	+2.5 dB	0.7	adds missing treble

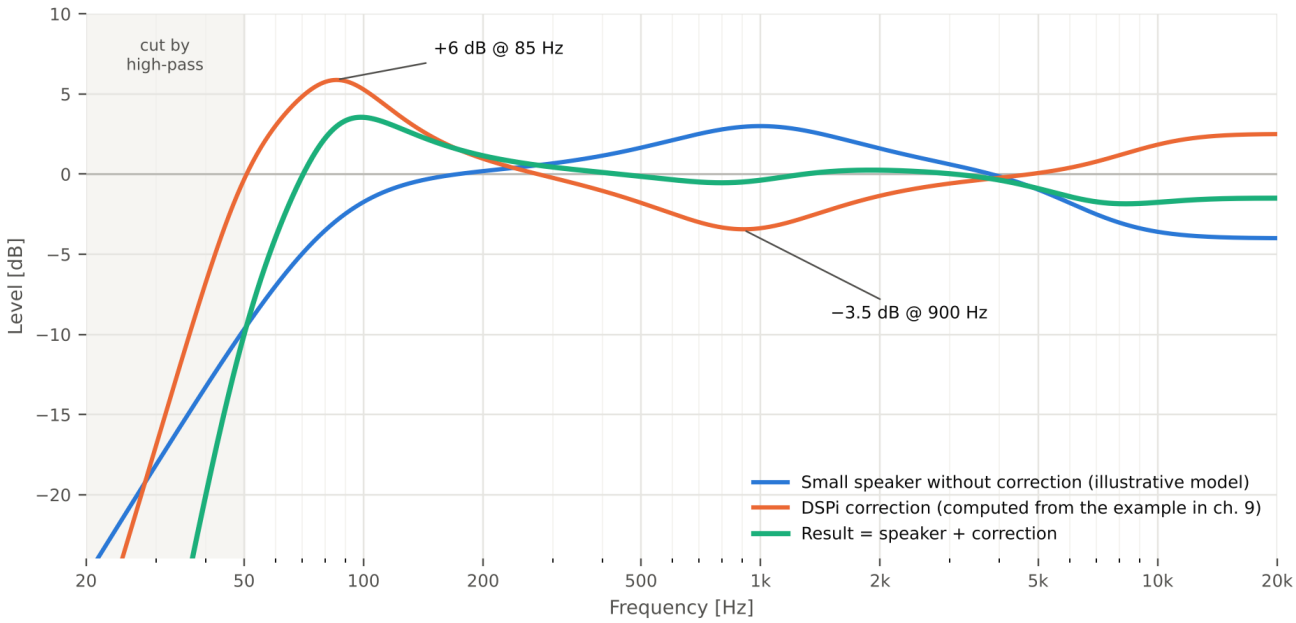
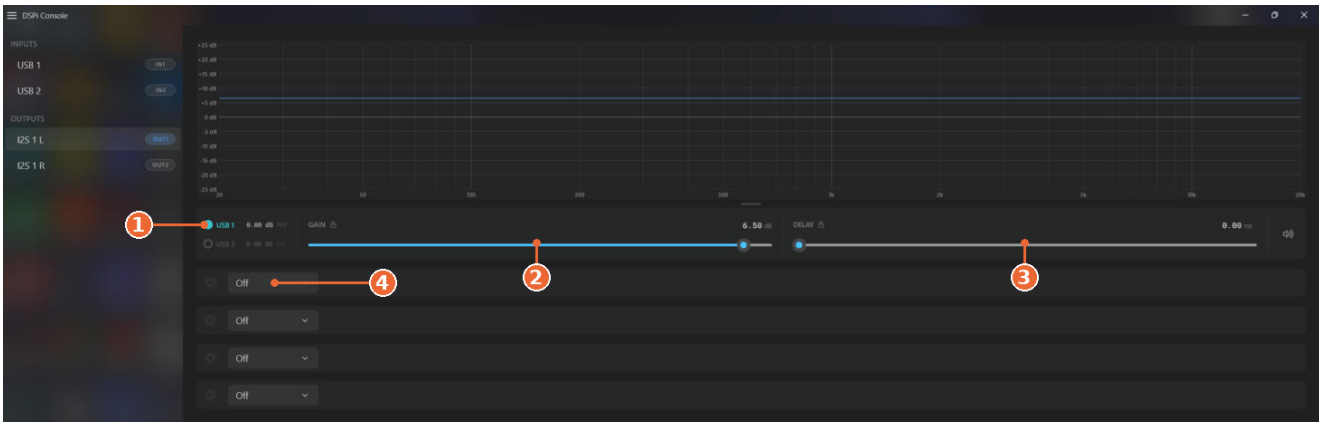


Fig. 4 – The correction curve (orange) is computed exactly from the filters in the table. The blue curve is an illustrative model of a mid-heavy small speaker, not a measurement. The green one shows how the correction adds up with such a curve.

If your speakers sound different – e.g. they have exaggerated bass instead – proceed the same way with opposite signs. Move the high-pass limit according to driver size: 60–70 Hz for 3" drivers, 35–45 Hz for 5–6" drivers. The high-pass belongs on the XO tab of the output channels, because DSPi offers steep filters up to 8th order there; the high-pass (Low Cut) in the PEQ is at most 2nd order (12 dB/oct). Enter PEQ filters in the table below the graph, or with the mouse directly in the graph: a double-click adds a band, dragging changes frequency and gain, the wheel changes the width (Q). On linked inputs (Link Pair) a change applies to both channels.



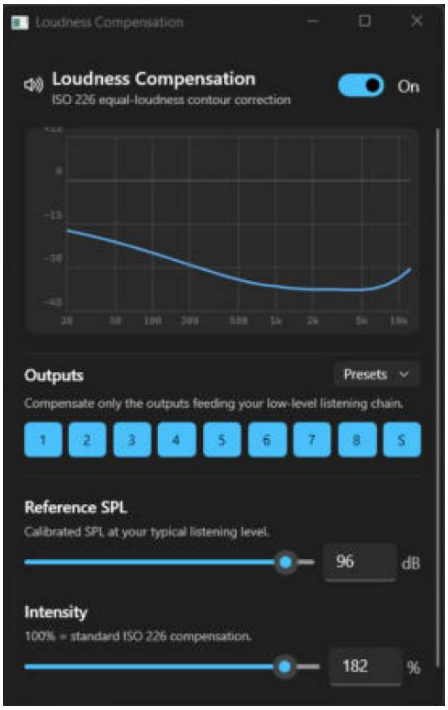
Screenshot 7 – Input channel editor: (1) channel USB 1, (2) Link Pair, (3) Preamp, (4) band filter type, (5) frequency, Q and gain, (6) Bypass All to compare without correction, (7) PEQ / XO tabs. The screenshot shows a real – rather extreme – correction of cheap desktop speakers, Preamp −26 dB.



Screenshot 8 – Output channel, XO tab: (1) signal source, (2) gain and (3) delay of the output, (4) up to four crossover filters (high-pass, low-pass, Linkwitz-Riley, Butterworth, Bessel).

9.4 Loudness

The human ear is less sensitive to bass and treble at low volume. Loudness in DSPi per ISO 226:2003 adds bass and treble the more, the lower the system volume. That is why in ch. 9.2 the volume is controlled in the system, not with the knob.



Screenshot 9 – Loudness Compensation (example settings; defaults are 83 dB and 100 %).

- Open the window with a **right click** on the Loudness icon in the toolbar (third from the left); a left click only toggles it.
- Set **Reference SPL** (default 83 dB) to the volume at which the system sounds balanced – typically your usual louder listening level.
- Reduce **Intensity** (default 100 % = standard ISO 226 curve) to 50–80 % if the correction is too strong at low volume.
- Loudness adds gain at low volume, where headroom is large; with the chosen Preamp it will not clip.
- The graph shows the correction added at reduced volume.

9.5 Psychoacoustic bass (optional)

Since version 1.1.5 DSPi can generate higher harmonics of bass tones for each output. The brain "reconstructs" the fundamental that a small driver physically cannot play. If your app version offers the function, try a gentle setting and possibly reduce the bass boost by 1–2 dB – it relieves the driver.

9.6 Listening procedure

1. Tune at your normal listening volume, at the place where you usually sit.
2. Use 3–4 well-known tracks of different genres and spoken word.
3. Change one parameter at a time, in small steps (1–2 dB, Q by 0.2).
4. Keep comparing with no correction (EQ bypass) – after a while the ear adapts and stops noticing the difference.
5. Watch the clip meters during loud passages.
6. Save the finished setting into a preset. In 10 slots you can store variants, e.g. "Music", "Night" (stronger loudness, smaller bass boost) and "Movie".

9.7 Tuning by measurement (advanced)

The ear is easily fooled. It is more accurate to measure the real frequency response at the listening position with a measurement microphone and acoustic measurement software (e.g. the free Room EQ Wizard) and design the filters from the measured curve. The parametric EQ in DSPi has enough bands and precision for that.

10. Presets and a control panel

So far everything was set in the app. DSPi can store up to 10 complete settings – **presets** – and switch them without a computer: with a button, a rotary knob or a remote control. This chapter shows how to turn one box into a switch between speakers and headphones with an "off" button, and what else a control panel can do.

10.1 What a preset contains

A preset is a complete snapshot of the processing settings. It is stored in the Pico memory and survives power loss.

- **Always:** EQ of all channels, preamp, EQ bypass, delays, loudness, crossfeed, volume leveller, the whole matrix (routing, phase inversion, gain) and the **enable and mute state of each output**, channel names.
- **Optionally:** master volume and output configuration (I2S/S/PDIF type, pins). By default they are global – switching presets does not change volume or wiring. The app can store them with each preset instead.
- **Switching** takes a moment; the output is muted for a few milliseconds, so it does not pop.
- **At power-on** either a chosen preset or the last active one is loaded.

10.2 A second output for headphones

The PCM5102A has a line output intended for high-impedance inputs; it cannot drive headphones well on its own. For headphones add a second DAC module on output slot 1 followed by a headphone amplifier. Both converters share the BCK and LRCK clocks and differ only in the data wire.

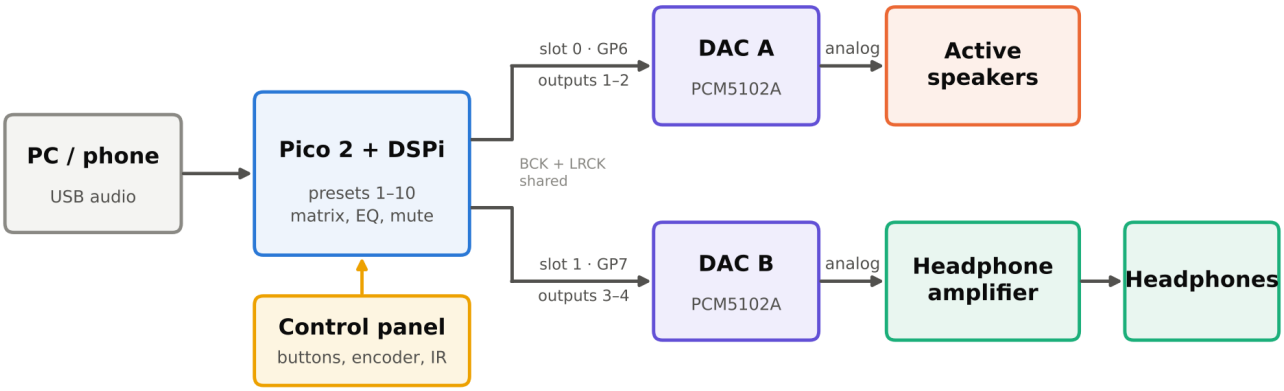


Fig. 5 – Two outputs from one Pico. The active preset determines which one plays.

DAC module B	Pico 2	Note
VIN	pin 40 / VBUS	shared with DAC A
GND	pin 38 / GND	shared with DAC A
BCK	pin 19 / GP14	shared with DAC A
LCK (LRCK)	pin 20 / GP15	shared with DAC A
DIN	**pin 10 / GP7**	separate – data of output slot 1

Split the shared wires with a soldered branch or on a breadboard – only one female connector fits on a Pico pin. Set the configuration inputs of DAC B the same as on DAC A (ch. 6). Then in DSPi Console:

1. switch **output slot 1** to I2S,
2. in the matrix route input L → output 3 and input R → output 4,
3. make sure the **PDM subwoofer output is off** – when it is on, only outputs 1–2 are available,
4. save the settings.

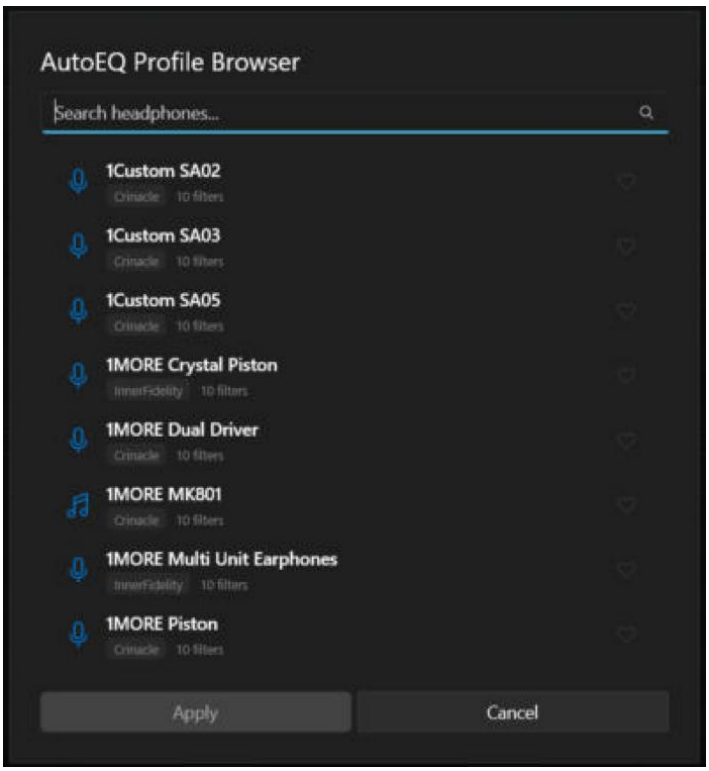
10.3 Example preset set

The mode is switched by each preset enabling different outputs and carrying its own correction. Recommended starting set:

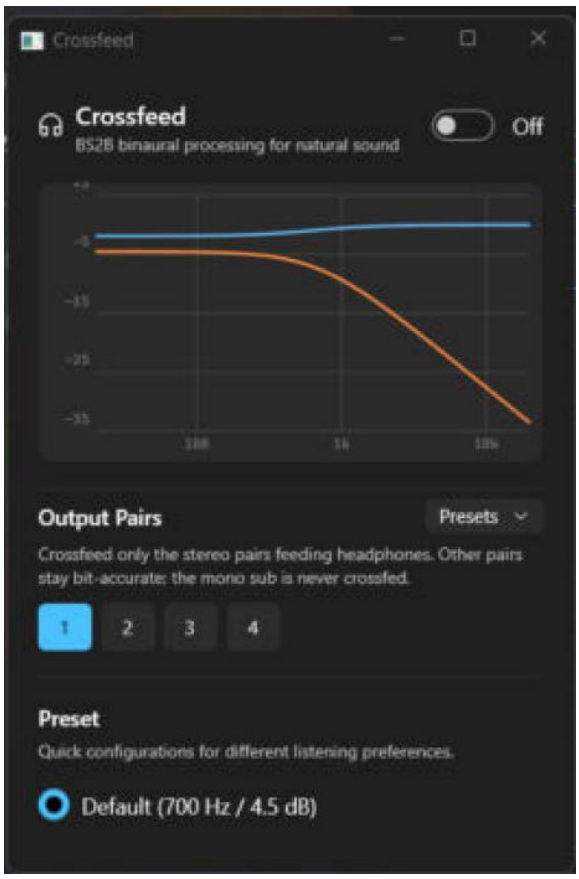
Setting	1 · Speakers	2 · Headphones	3 · Night	4 · Off
Outputs 1–2 (speakers)	● on	○ muted	● on	○ muted
Outputs 3–4 (headphones)	○ muted	● on	○ muted	○ muted
Master EQ	speaker correction (ch. 9.3)	headphone correction (AutoEQ)	like 1, less bass boost	—
Preamp	per the biggest boost	per AutoEQ	per the biggest boost	—
Crossfeed	off	on	off	—
Loudness	on	to taste	on, higher intensity	—
Volume leveller	off	off	on	—

- **Speakers** – correction and loudness per chapter 9.
- **Headphones** – DSPi Console includes **AutoEQ** with a database of more than a thousand headphone measurements; it loads the correction for your model in one step, including the recommended preamp. Crossfeed softens the unnaturally sharp stereo. Headphone volume is controlled by the amplifier knob.
- **Night** – the volume leveller boosts quiet passages (movie dialogue) and leaves loud ones alone; less bass boost spares the neighbours (ch. 10.6).
- **Off** – all outputs muted, both converters output silence.

Settings for the Headphones preset



Screenshot 10 – AutoEQ Profile Browser: searching for a headphone model.

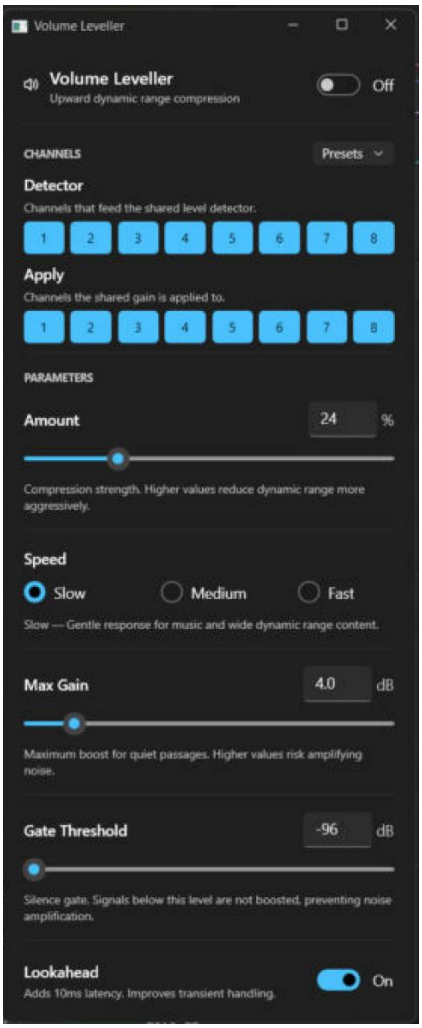


Screenshot 11 – Crossfeed: output pair selection and presets.

1. In inputs USB 1/2 clear the speaker correction (*Clear Filters* in the channel editor).
2. Open ☰ → **AutoEQ → Browse Profiles…**, search for your model and confirm with **Apply**. The app asks which channels to load the filters into – choose USB 1 and USB 2. It sets the profile's recommended preamp itself.
3. Right-click the Crossfeed icon (fourth from the left) to open its settings and choose a preset – *Default* (700 Hz / 4.5 dB) is a good start; a left click turns it on.
4. In the Matrix Mixer mute outputs OUT 1–2 and enable OUT 3–4. Save into slot 2.

Frequently used models can be saved to *AutoEQ → Favorites*; *Update Database…* refreshes the profile database.

Settings for the Night preset: volume leveller



Screenshot 12 – Volume Leveller (example of a gentle setting for music).

The leveller is an "upward" compressor: it boosts quiet passages towards a target level and leaves loud ones unchanged. Open its window with a right click on the speaker icon (sixth from the left). Starting points for night listening:

- **Amount** 50–70 % – strength of levelling (default 50 %).
- **Speed** Medium – Slow is gentle for music, Fast reacts quickly to dialogue.
- **Max Gain** 10–12 dB – maximum boost of quiet parts (default 15 dB).
- **Gate Threshold** around −50 to −60 dB – signal below the threshold is not boosted, so noise between tracks is not pulled up (the default −96 dB boosts almost everything).
- **Lookahead** on – adds 10 ms delay and smooths the reaction to sudden sounds.

These values are a recommended starting point, not factory settings. Fine-tune them by ear.

Saving and loading presets

Set up the mode, choose a slot in the preset selector at the bottom of the main window and save it with ☰ → **Save Preset** (Ctrl+S). Then in *Settings* → *General* → *Globals* choose **Startup mode**: *Last used* (loads the last used preset) or *Default preset* with **Default startup preset** – e.g. "Off" if you want to be sure that nothing plays at power-on. If speakers and headphones differ in sensitivity, balance them with the output gains (stored in the preset), or switch **Master volume mode** to *Per preset*.

10.4 Control panel: buttons, encoder and remote control

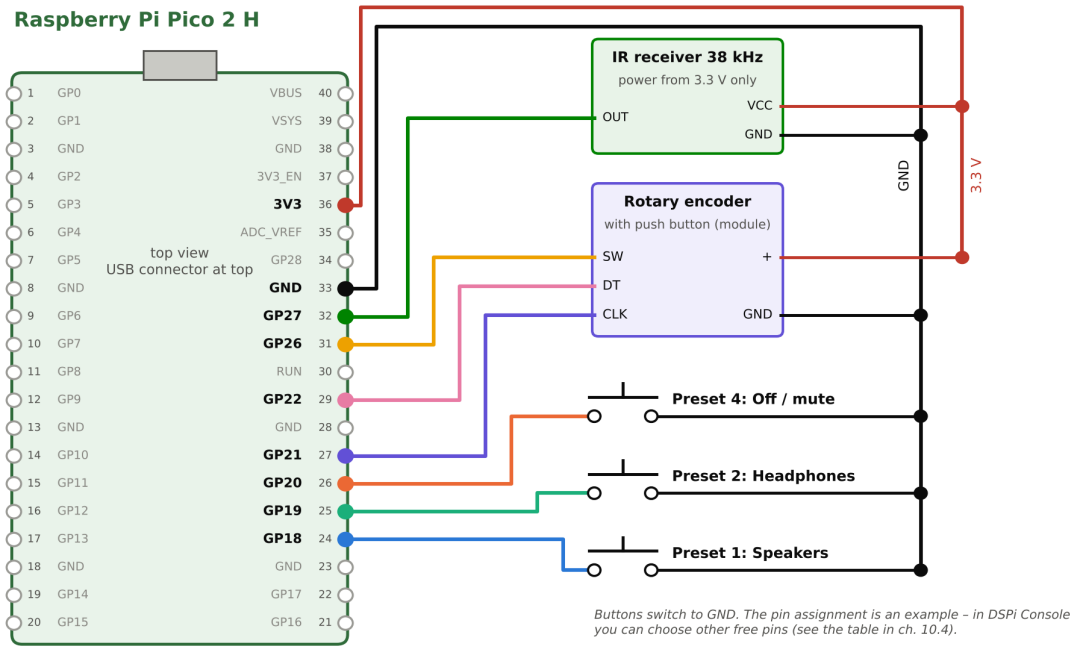


Fig. 6 – Example control panel: three preset buttons, a rotary encoder with push button and an IR receiver. The audio wires from ch. 6 and 10.2 are omitted for clarity.

The **Control Surfaces** feature (stable since 1.1.5) assigns DSP parameters to controls on free pins. It is set up in DSPi Console without programming and the configuration is stored in the Pico. Example assignment for fig. 6:

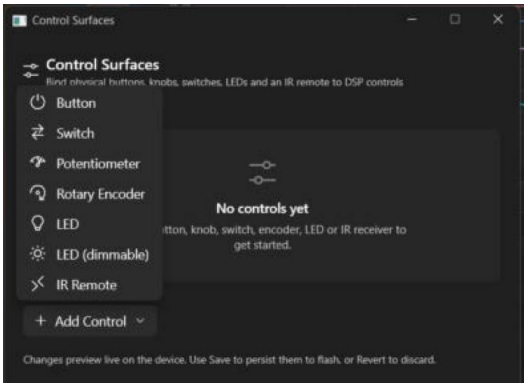
Control	Pico pin	Action	Target
Button 1	GP18 (pin 24)	set	preset 1 – Speakers
Button 2	GP19 (pin 25)	set	preset 2 – Headphones
Button 3	GP20 (pin 26)	set / toggle	preset 4 – Off, or mute
Encoder – rotation	GP21 CLK (pin 27), GP22 DT (pin 29)	step + / −	User Volume
Encoder – press	GP26 (pin 31)	toggle; long press	User Mute; preset 3 – Night
IR receiver	GP27 (pin 32)	learn remote buttons	volume ±, mute, presets 1–4
IR and encoder power	3V3 (pin 36), GND (pin 33)	—	—

What Control Surfaces offers

- **Controls:** buttons, switches, potentiometers, rotary encoders, LEDs (also PWM-dimmable) and IR receivers.
- **Actions:** absolute setting (potentiometer), step up/down, toggle, set value, momentary action (only while pressed), trigger and LED indication. Buttons distinguish long press, double press and auto-repeat.
- **Targets:** volume, mute, preset selection, input selection, enabling processing blocks (loudness, crossfeed…), output gain and delay and even individual filter parameters.
- **Remote control:** a learning mode captures NEC, RC5 and RC6 codes – i.e. most common remotes; up to 16 buttons.

Pin choice. Free pins are GP0, GP18–GP22 and GP26–GP28. The DSPi defaults use GP1–GP5 (I2S and S/PDIF inputs), GP6–GP9 (outputs), GP10 (PDM subwoofer), GP11 (DAC hardware mute), GP13 (MCK), GP14–GP15 (I2S clocks) and GP16–GP17 (debug UART); GP23–GP25 are reserved by the board. Unused functions can be moved in the app, but the simplest is to stick to the free pins.

Procedure in the app (version 1.1.5)



Screenshot 13 – Control Surfaces: the Add Control menu.

The window is empty at first. Add each control with **Add Control** and the control type; changes take effect on the device immediately and Save stores them in the Pico memory.

1. Open ☰ → **Control Surfaces…** (Ctrl+Shift+I) and click **Add Control**.
2. Choose the control type: *Button*, *Switch*, *Potentiometer*, *Rotary Encoder*, *LED*, *LED (dimmable)* or *IR Remote*, and the pin – two pins *GPIO A* and *GPIO B* for an encoder.
3. For buttons switching to GND leave **Active-low input (pull-up)** on – the Pico uses its internal pull-up resistor, no extra parts are needed.
4. Choose the event (*Press*, *Long press*, *Double press*), the action (e.g. *Set value*, *Toggle*, *Step*) and the parameter: **Preset** with values *Preset 1* to *Preset 10* for presets, **User Volume** for volume, **User Mute** for mute.
5. Confirm with **Apply** – changes take effect on the device right away – and test. Finally **Save** stores the configuration in the Pico memory, **Revert** discards it.
6. Remote control: add an *IR Remote* control with the receiver pin and Apply. Then **Add Remote Button → Learn Button**, press the button on the remote and assign an action to it as for a button.

Caution
User Volume is the same volume the operating system sets, and loudness follows it (ch. 9.4). **Master Volume** is the device ceiling – loudness does not follow it. So use User Volume for the encoder. Power the IR receiver and encoder module from 3.3 V (pin 36), not 5 V – the Pico inputs do not tolerate 5 V.

10.5 Complete silence and switching off

- **Mute button or Off preset?** Mute is quick and restores exactly the previous mode when released. The Off preset can be set as the power-on default. You can have both (button 3 and encoder press in the table above).
- **Hardware mute of the converter.** The External Mute Control function drives the XSMT pin of the converter from the Pico (default GP11), so the output is truly silent – even while switching and saving settings. It is set in *Settings → Hardware → External Mute Control*: *Enable*, *GPIO Pin*, *Polarity* = **Active Low** (for the PCM5102A), and *Begin Test* verifies the wiring. Only if your module brings out XSMT on a separate pin; its jumper must then not be hard-wired to H or L. If unsure, skip this option.

- **Physically switching the speakers off.** Auxiliary outputs (version 1.1.6-beta3) can drive a relay or an amplifier trigger input, so the Off preset can also disconnect the speakers from power.

> **Safety**
> Switch relays only on the low-voltage side (DC after the power adapter). Switching mains voltage belongs only to qualified persons – for mains appliances a ready-made smart plug is safer.

10.6 More control and processing options

Features in version 1.1.5 are stable. Newer ones are in beta releases so far – for reliable operation stay with the stable release, or expect occasional bugs. The app and firmware versions must match.

Feature	Version	What it is for
Volume leveller	1.1.5	"upward" compressor: boosts quiet passages to a target level, leaves loud ones unchanged. Useful for movies and night listening (settings see Screenshot 12); it does not limit peaks.
Output limiter	1.1.6-beta4	a "brickwall" limiter on each output – stops peaks above the set threshold and protects speakers from overload
Input selection	1.1.5	switch USB, S/PDIF or I2S with a button – e.g. computer and TV
Test signal generator	1.1.5	15 signal types and a short "chirp" to identify which output is which
Control over UART and I2C	1.1.5	control from another microcontroller (e.g. ESP32) without USB
Macros	1.1.6-beta1	one button runs a sequence of actions: select an input and load a preset for it, A/B switching, timed steps
Target groups	1.1.6-beta1	one knob controls a group of channels and keeps their balance
Display	1.1.6-beta2	character (HD44780 with PCF8574 backpack, US2066) or graphic (SSD1306, SH1106); shows the parameter being changed or cycles up to 16 pages, editing with a single encoder
Auxiliary outputs	1.1.6-beta3	driving relays, amplifier trigger inputs or lighting
Spectrum analyser	1.1.6-beta3	1024-point FFT on an input or output

11. Troubleshooting

Symptom	Likely cause	Fix
The Pico does not appear as an audio device	charge-only cable; wrong .uf2 file	use a data cable; flash the RP2350 file
The device is visible but nothing plays	slot 0 still S/PDIF; BCK/LCK/DIN swapped; XSMT at L	switch slot 0 to I2S and save; check wiring; XSMT to H
Silence again after re-plugging	the settings were not saved	save the preset / output config
Heavy distortion or loud noise	FMT at H or SCK not connected	FMT to L, SCK to GND
Distortion only in loud passages	digital clipping	increase the Preamp attenuation (ch. 9.2)
Hum or whine	ground loop computer – speakers	USB isolator or a different USB port
Occasional pops or dropouts	poor cable, USB hub; long wires between modules	plug directly into the PC, shorter cables
Sound stops after a while, meters still move	known with some beta firmware; re-opening the audio stream helps	use the stable firmware; switch the Windows output away and back as a workaround
Very quiet sound	master volume left at −20 dB	set 0 dB (ch. 9.2)
Outputs 3–4 (headphones) silent	slot 1 not I2S; PDM subwoofer on; outputs muted in the preset	switch slot 1 to I2S, turn PDM off, check mute in the matrix (ch. 10.2)
Button or IR does not react	pin used by another function or not assigned; IR powered from 5 V	choose a free pin and assign it in Control Surfaces; IR on 3.3 V (ch. 10.4)
The app does not connect or behaves oddly	different DSPi Console and firmware versions	match the versions (ch. 8)
The app reports that libusb sees no device	the control interface has no WinUSB driver	assign WinUSB with Zadig (ch. 8.3)
S/PDIF and silence again after unplugging	the output configuration was not saved	☰ → Save Output Config (ch. 8.4)

Where to find help
The project community is active on Discord (link in the repository README) and in GitHub discussions.

12. Where to go next

The described setup uses only a fraction of what DSPi can do. You can build on the same base:

- **Subwoofer.** The dedicated PDM output needs only a simple RC filter; crossover frequency and delay are set in the app. Alternatively a second DAC module on another output slot.
- **Active crossover.** The Pico 2 has up to 8 output channels – enough for a stereo system with up to four ways, each with its own amplifier.
- **Analogue input.** A PCM1808 A/D converter module connected via I2S lets you process audio from a TV or turntable.
- **Optical S/PDIF input.** With a TORX141-type receiver you can connect a TV or game console.
- **Network or home-automation control.** Over UART or I2C another microcontroller can control DSPi, e.g. an ESP32 with Wi-Fi (ch. 10.6).

13. Links

DSPi – source code and documentation
https://github.com/WeebLabs/DSPi

DSPi – firmware releases (.uf2 files)
https://github.com/WeebLabs/DSPi/releases

DSPi – I2S output specification
https://github.com/WeebLabs/DSPi/blob/main/Documentation/Features/i2s_output_spec.md

Weeb Labs – DSPi Console, DSPi Terminal and more
https://github.com/WeebLabs

DSPi Console for Windows – documentation
https://github.com/WeebLabs/DSPi-Console-Windows

DSPi Console for Windows – download
https://github.com/WeebLabs/DSPi-Console-Windows/releases

Raspberry Pi Pico 2 – product page
https://www.raspberrypi.com/products/raspberry-pi-pico-2/

Raspberry Pi Pico 2 pinout (PDF)
https://datasheets.raspberrypi.com/pico/Pico-2-Pinout.pdf